

DU OUYANG

CONTACT INFORMATION

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EDUCATION

Department of Mathematical Sciences, Tsinghua University Sep. 2021 – Jun. 2026
Ph.D. in Mathematics, with Distinction Beijing, China
Supervisor: Prof. Xiaoqun Wang, Tsinghua University

School of Mathematics and Statistics, Wuhan University Sep. 2017 – Jun. 2021
Bachelor of Mathematics and Applied Mathematics Wuhan, China

VISITING EXPERIENCES

The Chinese University of Hong Kong Sep. 2024 – Mar. 2025, Jul. 2025 – Feb. 2026
Visiting Ph.D. Student (Advisor: Prof. Yanwei Jia) Hong Kong

- Conducting research on continuous-time reinforcement learning.

RESEARCH INTERESTS

- Numerical Analysis
- Stochastic Control
- Reinforcement Learning
- Machine Learning

PUBLISHED PAPERS

- **Du Ouyang**, Xiaoqun Wang, and Zhijian He, *Achieving high convergence rates by quasi-Monte Carlo and importance sampling for unbounded integrands*, **SIAM Journal on Numerical Analysis**, 62 (2024), pp. 2393-2414.
- **Du Ouyang**, Jichang Xiao, and Xiaoqun Wang, *Generalization error analysis of deep backward dynamic programming for solving nonlinear PDEs*, **Applied Numerical Mathematics**, 217 (2025), pp. 319-339.
- Yanwei Jia, **Du Ouyang**, and Yufei Zhang, *Accuracy of discretely sampled stochastic policies in continuous-time reinforcement learning*, **SIAM Journal on Control and Optimization**, 64 (2026), pp. 1889-1929. (Alphabetical order)

WORKING PAPERS

- Zexin Pan, **Du Ouyang**, and Zhijian He, *Quasi-Monte Carlo integration over $\mathbb{R}^s$ with boundary-damping importance sampling*, arXiv preprint arXiv:2509.07509 (2025), under review at SIAM Review.

- Zexin Pan, **Du Ouyang**, and Zhijian He, *Uncertainty quantification using importance-sampled quasi-Monte Carlo with dimension-independent convergence rates*, arXiv preprint arXiv:2603.00843 (2026), under review at SIAM Journal on Numerical Analysis.
- Yanwei Jia, **Du Ouyang**, Huyên Pham, and Xunyu Zhou, *A zeroth-order deep learning method for fully nonlinear parabolic partial differential equations*, arXiv preprint arXiv:2606.24999 (2026), submitted to Journal of Machine Learning Research. (*Alphabetical order*)
- **Du Ouyang**, Zexin Pan, and Zhijian He, *Quasi-Monte Carlo for SDE simulation: Error analysis and dimensionality reduction*, arXiv preprint arXiv:2606.24207 (2026).
- **Du Ouyang**, Zhenpeng Shi, Aodong Guo, Huaze Tang, Hejin Wang, Chao Wang, Wenbo Ding, *Adaptive Two-Level Quasi-Monte Carlo for Soft Actor-Critic*, ICML 2024 Workshop: Aligning Reinforcement Learning Experimentalists and Theorists.

ACADEMIC PRESENTATIONS

- **Recent Advances in the quasi-Monte Carlo Methods**, South China University of Technology, September 2023.
- **The 15th POMS-HK International Conference**, The Chinese University of Hong Kong, January 2025
- **The 9th PKU-NUS Annual International Conference**, Peking University, May 2025
- **The 15th International Conference on Monte Carlo Methods and Applications (MCM 2025)**, Illinois Institute of Technology, Chicago, August 2025
- **17th International Conference on Monte Carlo and Quasi-Monte Carlo Methods in Scientific Computing (MCQMC 2026)**, The University of Edinburgh, Edinburgh, June 2026

EXPERIENCES

- **Teaching Assistant**
 - Calculus, Spring 2022 and Fall 2022.
 - Probability Theory and Mathematical Statistics, Fall 2021 and Fall 2023.
 - Probability Theory, Spring 2023.
- **Captain of the Football Team of Department of Mathematical Sciences**, Tsinghua University, 2023.

AWARDS

- **National Scholarship for Doctoral Students**, Tsinghua University, 2024.
- **THU Postgraduate Excellency Scholarship First Prize**, Tsinghua University, 2023, 2025.
- **Excellent TA Award**, Department of Mathematical Sciences, Tsinghua University, 2023.
- **Outstanding Ph.D. Graduate (Top 5%)**, Tsinghua University, 2026
- **Outstanding Doctoral Dissertation Award**, Tsinghua University, 2026